

HDOC DAY-6 ACTIVITY

C Programming Assignment

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Question 1: Radius, Area and Perimeter of a Semicircle

Step 1: Problem Understanding

Problem Statement: Given the diameter of a semicircle, find its radius, area, and perimeter.

- Input: Diameter of the semicircle (a floating-point value).
- Output: Radius, Area, and Perimeter of the semicircle.

Formulas used:

- Radius (r) = Diameter / 2
- Area = $(\pi \times r^2) / 2$
- Perimeter = $\pi \times r$ + Diameter (curved boundary + straight diameter edge)

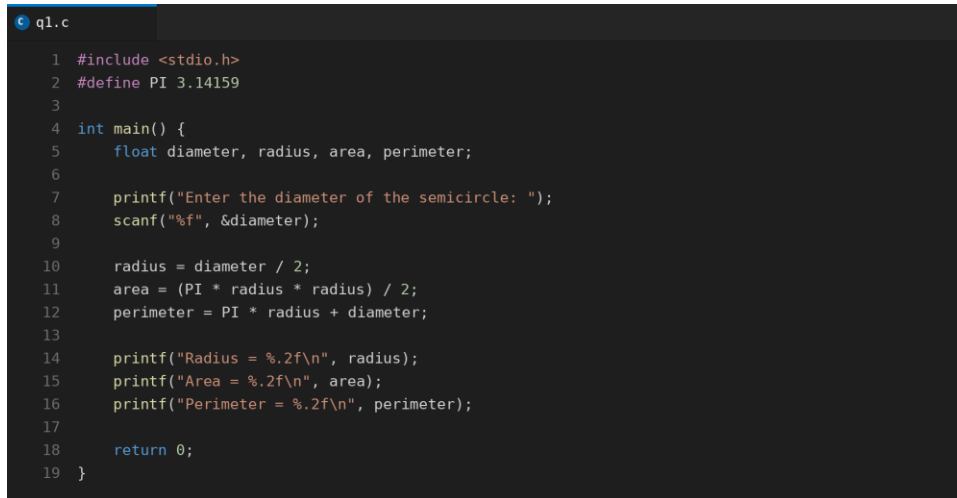
Step 2: Test Case Table

Test Case	Input	Expected Output	Actual Output	Result (Pass/Fail)
1	diameter = 10	Radius = 5.00 Area = 39.27 Perimeter = 25.71	Radius = 5.00 Area = 39.27 Perimeter = 25.71	Pass
2	diameter = 14	Radius = 7.00 Area = 76.97 Perimeter = 35.99	Radius = 7.00 Area = 76.97 Perimeter = 35.99	Pass
3	diameter = 0	Radius = 0.00 Area = 0.00 Perimeter = 0.00	Radius = 0.00 Area = 0.00 Perimeter = 0.00	Pass

Step 3: Algorithm

- Step 1: Start.
- Step 2: Declare variables diameter, radius, area, and perimeter as floating-point numbers.
- Step 3: Read the value of diameter from the user.
- Step 4: Compute radius = diameter / 2.
- Step 5: Compute area = $(\pi \times \text{radius} \times \text{radius}) / 2$.
- Step 6: Compute perimeter = $(\pi \times \text{radius}) + \text{diameter}$.
- Step 7: Display radius, area, and perimeter.
- Step 8: Stop.

Step 4: C Program

A screenshot of a VS Code editor window showing a C program named q1.c. The code calculates the radius, area, and perimeter of a semicircle based on a user-provided diameter. The code is as follows:

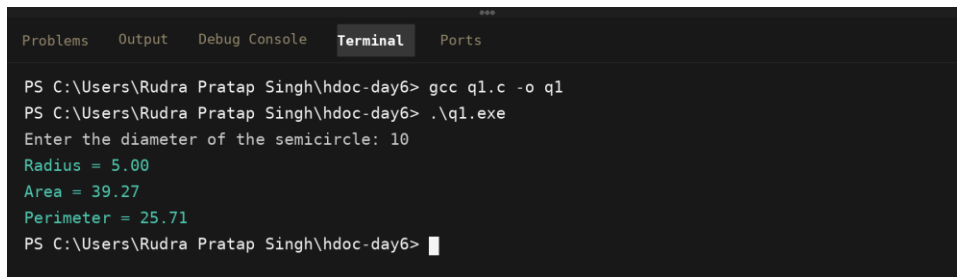
```
1 #include <stdio.h>
2 #define PI 3.14159
3
4 int main() {
5     float diameter, radius, area, perimeter;
6
7     printf("Enter the diameter of the semicircle: ");
8     scanf("%f", &diameter);
9
10    radius = diameter / 2;
11    area = (PI * radius * radius) / 2;
12    perimeter = PI * radius + diameter;
13
14    printf("Radius = %.2f\n", radius);
15    printf("Area = %.2f\n", area);
16    printf("Perimeter = %.2f\n", perimeter);
17
18    return 0;
19 }
```

Fig 1.1 — q1.c in VS Code editor

Step 5: Compilation, Execution and Evaluation

The program was compiled using GCC (`gcc q1.c -o q1`) with no errors or warnings, and executed successfully for all three test cases listed in Step 2. In each case the Actual Output produced by the program matched the Expected Output exactly, so all test cases are marked Pass.

Step 6: Compilation and Execution Snippet

A screenshot of the VS Code terminal window showing the compilation and execution of the q1.c program. The terminal output is as follows:

```
PS C:\Users\Rudra Pratap Singh\hdoc-day6> gcc q1.c -o q1
PS C:\Users\Rudra Pratap Singh\hdoc-day6> .\q1.exe
Enter the diameter of the semicircle: 10
Radius = 5.00
Area = 39.27
Perimeter = 25.71
PS C:\Users\Rudra Pratap Singh\hdoc-day6> 
```

Fig 1.2 — Compilation and execution of q1.c in the VS Code terminal (diameter = 10)

Question 2: Total Area of a Rectangle with a Triangular Top

Step 1: Problem Understanding

Problem Statement: A shape is formed by a rectangle with a triangle placed on top of it (like a simple house shape). Given the rectangle's length and breadth, and the triangle's base and height, find the total area of the combined shape.

- Input: Length and breadth of the rectangle; base and height of the triangle.
- Output: Area of the rectangle, area of the triangle, and the total combined area.

Formulas used:

- Rectangle Area = length $\times$ breadth
- Triangle Area = $0.5 \times$ base $\times$ height
- Total Area = Rectangle Area + Triangle Area

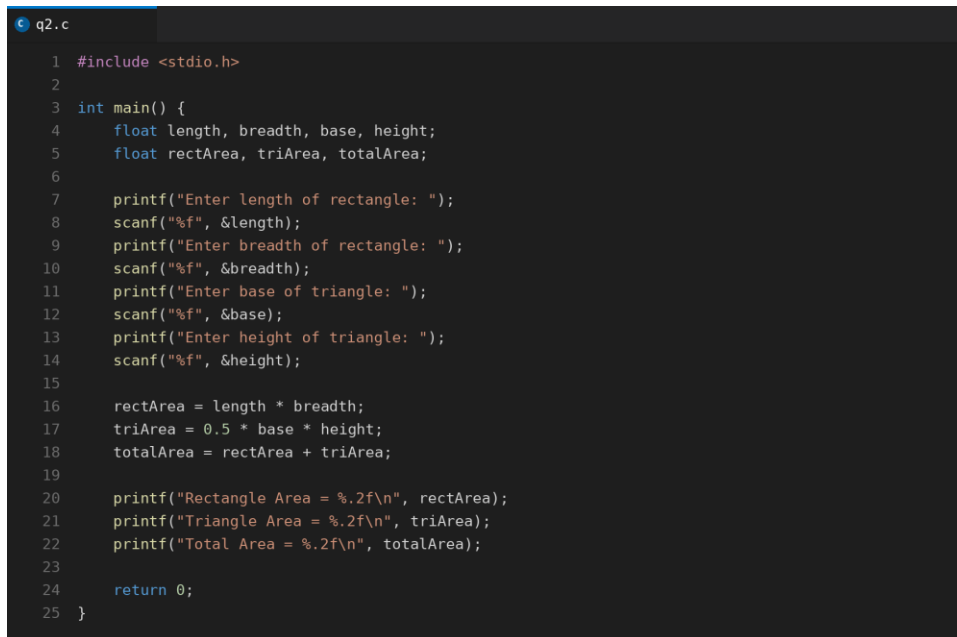
Step 2: Test Case Table

Test Case	Input	Expected Output	Actual Output	Result (Pass/Fail)
1	length = 6, breadth = 4, base = 6, height = 3	Rectangle Area = 24.00 Triangle Area = 9.00 Total Area = 33.00	Rectangle Area = 24.00 Triangle Area = 9.00 Total Area = 33.00	Pass
2	length = 10, breadth = 5, base = 8, height = 4	Rectangle Area = 50.00 Triangle Area = 16.00 Total Area = 66.00	Rectangle Area = 50.00 Triangle Area = 16.00 Total Area = 66.00	Pass
3	length = 0, breadth = 0, base = 0, height = 0	Rectangle Area = 0.00 Triangle Area = 0.00 Total Area = 0.00	Rectangle Area = 0.00 Triangle Area = 0.00 Total Area = 0.00	Pass

Step 3: Algorithm

- Step 1: Start.
- Step 2: Declare variables length, breadth, base, height, rectArea, triArea, and totalArea as floating-point numbers.
- Step 3: Read length and breadth of the rectangle from the user.
- Step 4: Read base and height of the triangle from the user.
- Step 5: Compute rectArea = length $\times$ breadth.
- Step 6: Compute triArea = $0.5 \times$ base $\times$ height.
- Step 7: Compute totalArea = rectArea + triArea.
- Step 8: Display rectArea, triArea, and totalArea.
- Step 9: Stop.

Step 4: C Program



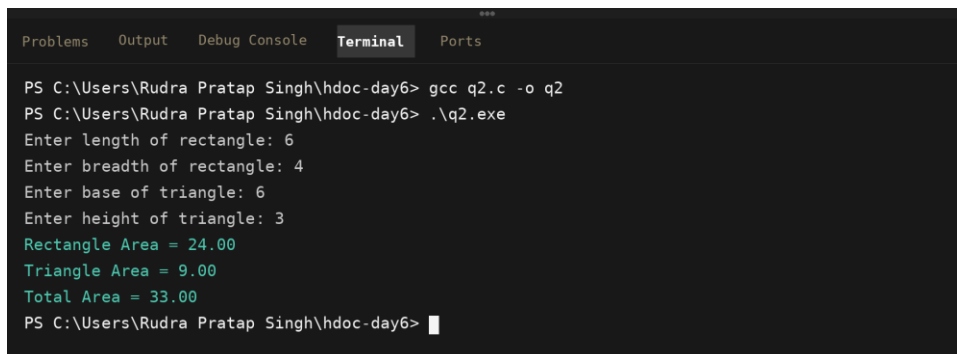
```
1 #include <stdio.h>
2
3 int main() {
4     float length, breadth, base, height;
5     float rectArea, triArea, totalArea;
6
7     printf("Enter length of rectangle: ");
8     scanf("%f", &length);
9     printf("Enter breadth of rectangle: ");
10    scanf("%f", &breadth);
11    printf("Enter base of triangle: ");
12    scanf("%f", &base);
13    printf("Enter height of triangle: ");
14    scanf("%f", &height);
15
16    rectArea = length * breadth;
17    triArea = 0.5 * base * height;
18    totalArea = rectArea + triArea;
19
20    printf("Rectangle Area = %.2f\n", rectArea);
21    printf("Triangle Area = %.2f\n", triArea);
22    printf("Total Area = %.2f\n", totalArea);
23
24    return 0;
25 }
```

Fig 2.1 — q2.c in VS Code editor

Step 5: Compilation, Execution and Evaluation

The program was compiled using GCC (`gcc q2.c -o q2`) with no errors or warnings, and executed successfully for all three test cases listed in Step 2. In each case the Actual Output produced by the program matched the Expected Output exactly, so all test cases are marked Pass.

Step 6: Compilation and Execution Snippet



```
Problems Output Debug Console Terminal Ports
PS C:\Users\Rudra Pratap Singh\hdoc-day6> gcc q2.c -o q2
PS C:\Users\Rudra Pratap Singh\hdoc-day6> .\q2.exe
Enter length of rectangle: 6
Enter breadth of rectangle: 4
Enter base of triangle: 6
Enter height of triangle: 3
Rectangle Area = 24.00
Triangle Area = 9.00
Total Area = 33.00
PS C:\Users\Rudra Pratap Singh\hdoc-day6> 
```

Fig 2.2 — Compilation and execution of q2.c in the VS Code terminal (length=6, breadth=4, base=6, height=3)